

Tianruo Rose Xu

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Education

Cornell University, College of Engineering

Aug 2024 – May 2027 (expected)

B.S. in Computer Science Minor in Mathematics

Selected coursework: Natural Language Processing, Deep Learning, Machine Learning, Reinforcement Learning, AI Practicum, Algorithms, Data Structures and Functional Programming, Computer Systems, Mathematical Foundations of Computing, Applied Logic, Linear Algebra, Probability and Statistics, Multivariable Calculus, ODEs, Number Theory, Mathematical Modeling.

Honors and Awards

- Hunter R. Rawlings III Cornell Presidential Research Scholar Award.
- Clare Boothe Luce Undergraduate Research Award.
- Bowers Undergraduate Research Experience (BURE) stipend and Academic Year Expansion support.
- USA Computing Olympiad (USACO) Platinum Division; perfect score in the [January 2023 Gold Contest](#).
- USA Mathematical Olympiad (USAMO) Qualifier.

Publications

[Submitted] *Can Emergent Verbalization of Epistemic Uncertainty Inform Factuality?*

[Published] **Tianruo Rose Xu**. *Can You Trust What I Think? Analyzing and Improving Verbalized Uncertainty and Factuality in Reasoning-Based Large Language Models*. AAAI-26 Undergraduate Consortium, Singapore, January 2026. [\[Poster\]](#) [\[Oral Presentation\]](#) [\[Paper\]](#).

[Published] **Tianruo Rose Xu**, Vedant Gaur, Liu Leqi, Tanya Goyal. *The Progress Illusion: Revisiting Meta-Evaluation Standards of LLM Evaluators*. Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, November 2025. [\[Paper\]](#) [\[Poster\]](#).

[Published] Yingke Ding, Jiankai Tang, Wanying Mo, **Tianruo Rose Xu**, Yuanchun Shi, Yuntao Wang. *NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model*. MICCAI 2025 Workshop MSB EMERGE, July 21, 2025. [\[Paper\]](#) [\[Code\]](#). Runner-up Best Paper, MICCAI 2025 Workshop MSB EMERGE.

[Published] Tianruo Xu. *Elder Depression Detection by Multimodal Means*. BCP Education & Psychology, November 7, 2022. [\[Paper\]](#).

Research Experience

Undergraduate Researcher, Cornell University

Computer Science Department, Natural Language Processing Group; advised by Professor Tanya Goyal.

Analyzing Verbalized Uncertainty and Factuality in Reasoning-Based LLMs 2025 – 2026

- Investigates whether verbalized epistemic uncertainty in LLM reasoning traces predicts fact-level factuality in long-form question answering.
- Led project design with Anirudh Atmakuru and Sanjana Ramprasad; built a pipeline for trace analysis, atomic fact extraction, trace-response alignment, FactScore evaluation, large-scale empirical analysis, and classifier training.

Revisiting Meta-Evaluation Standards of LLM Evaluators 2025

- Develops fine-grained meta-evaluation methods for auditing LLM-as-judge benchmarks, evaluator reliability, and reference-model sensitivity.
- Led the evaluation framework design; implemented data collection and scoring pipelines; ran large-scale experiments; analyzed correlations with human judgments; contributed to writing, figures, and result interpretation.

Content and Style Evaluation in Natural Language Generation Systems 2024

- Studies content and style as separable evaluation dimensions in natural language generation systems.
- Co-supervised by Professor Claire Cardie; designed experiments, implemented data processing and evaluation pipelines, ran model-based evaluations, conducted error analysis and ablations, and contributed to writing and result visualization.

Undergraduate Researcher, Cornell University

Mathematics Department, Math + AI Lab; advised by Professor Daniel Halpern-Leistner.

Lean Proof Arena and LaTeX-to-Lean Autoformalization Pipeline 2026 – present

- Develops a DARPA ExPMath-supported platform and pipeline for research-level autoformalization, centered on Lean Proof Arena and LaTeX-to-Lean conversion.
- Building Lean Proof Arena for comparing formal proof attempts; developing a LaTeX-to-Lean pipeline, benchmark tasks, and evaluation protocols for formal correctness and semantic fidelity.

Research Intern, UC San Diego

Computer Science Department, Q-Lab; advised by Professor Lianhui Qin and Professor Haojian Jin.

SimWorld Real-Time Safety Reasoning 2026 – present

- Studies real-time safety risks, dangerous behaviors, and failure modes of embodied AI agents in realistic SimWorld environments.
- Lead the project by defining the research direction, safety scenario taxonomy, benchmark tasks, and evaluation workflow for systematic analysis of embodied AI risks.

DeliveryBench V2: Embodied Delivery Agent Benchmark in SimWorld 2026 – present

- Extends DeliveryBench into a more robust SimWorld benchmark for evaluating embodied agents on long-horizon delivery tasks with realistic navigation, resource, and decision-making constraints.
- Collaborated on environment refinement, task design, simulation debugging, reinforcement learning training setup, experiment execution, and benchmark evaluation.

Undergraduate Research Assistant, Tsinghua University

Department of Computer Science and Technology, Pervasive Human-Computer Interaction Lab.

Intelligent AI Tool Benchmarking and Orchestration Platform 2025 – present

- Develops an intelligent AI tool platform that standardizes tool capability descriptions, matches tools to user intents, and supports multi-tool workflow orchestration.
- Designed and implemented back-end search and matching algorithms for tool retrieval, user-intent matching, and personalized AI tool selection.

NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model 2025

- Develops a multi-task learning model for automated nailfold capillaroscopy, combining segmentation, classification, and keypoint detection for non-invasive microvascular image analysis.
- Conducted dataset preprocessing, baseline training, experiment execution, model evaluation, and preliminary result analysis to support development of the NFCMTL framework.

Undergraduate Research Assistant, Cornell University 2024

Information Science Department; advised by Keigo Kusumegi and Professor Yian Yin.

- Studied how diverse collaboration patterns affect Wikipedia content quality.
- Processed Wikipedia data and applied K-means clustering to analyze article-quality patterns.

Teaching Experience

Cornell University

- **Head Teaching Assistant**, CS 3780: Introduction to Machine Learning Fall 2026
- **Teaching Assistant**, CS 3780: Introduction to Machine Learning Spring 2026
Designed the final Kaggle challenge, [All About Birds!](#), a bird-sound spectrogram classification task.
- **Teaching Assistant**, CS 4820: Introduction to Analysis of Algorithms Fall 2025
- **Teaching Assistant**, CS 2800: Mathematical Foundations of Computing Spring 2025